

On the Near Wake Turbulent Flow Properties of the SD7003 Airfoil

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Does the maximum aerodynamic efficiency condition correspond to reduced turbulent fluctuations in the near wake? Does approaching maximum aerodynamic efficiency effectuate onset of flow separation on the wing? This research is an endeavor to answer these fundamental questions and explore the relationship between near wake turbulent properties and aerodynamic efficiency. An SD7003 wall-to-wall model was chosen for this study. Time Resolved Particle Image Velocimetry (TR-PIV) was performed in the near wake of a wall-to-wall SD7003 wing for angles of attack (AoA) ranging from -2 to 8 degrees at AFRL's Horizontal Free-Surface Water Tunnel (HFWT). A detailed analysis of the near wake was performed to understand the trends between the mean (momentum deficit, vorticity, and two-point correlation) and fluctuating (U_{RMS} , V_{RMS} , Reynolds shear stress and? frequency spectrum) near wake flow quantities with aerodynamic efficiency. The results show evidence of three different wake signatures: 1) prior to the maximum aerodynamic efficiency with periodic vortex shedding, 2) in the vicinity of maximum aerodynamic efficiency with reduced magnitudes of mean and fluctuating velocity components in the wake and 3) beyond the maximum aerodynamic efficiency condition with an increase in the magnitudes of mean and fluctuating velocity components. The resulting correlations between the aerodynamic efficiency and the near wake properties could lead to better performance, attenuating unsteady aerodynamic loads, and even harvesting energy from the wake.

Nomenclature

L/D	=	lift-to-drag ratio
c	=	chord (m)
δ	=	displacement thickness (m)
ω	=	vorticity ($1/s$)
U_{∞}	=	freestream velocity (m/s)
u'	=	streamwise fluctuating velocity (m/s)
v'	=	transverse fluctuating velocity (m/s)
U_{RMS}	=	streamwise root-mean-square velocity (m/s)
$u'v'$	=	Reynolds shear stress (m^2/s^2)
Δt	=	time delay (s)
α	=	angle of attack ($degrees$)
ρ	=	two-point correlation
x, y, z	=	co-ordinate systems (m)
L_0	=	wake-half width (m)

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I. Introduction and Background

The maximum aerodynamic efficiency condition defined by the Lift-to-Drag ratio corresponds to a condition of minimum drag for a given quantity of lift produced. There are however, some lingering fundamental questions to explore such as

- Does approaching the maximum aerodynamic efficiency condition translate to inherent changes in near wake turbulent properties?
- Does the maximum aerodynamic efficiency condition effectuate onset of flow separation?
- Does the maximum aerodynamic efficiency condition correspond to minimum entropy generation?
- Does the maximum aerodynamic efficiency condition impact the degree of wake symmetry/asymmetry?
- Do the above-mentioned relationships remain consistent across a range of Reynolds numbers?

Due to the variety of Reynolds numbers and conditions that the new generation of drones and air vehicles operate across, it is now more important than ever to understand the relationship between the aerodynamic efficiency and turbulent flow properties in the near wake. Better understanding of this relationship could lead to increased aerodynamic performance, attenuating unsteady aerodynamic loads and alleviating structural vibrations. It will also benefit energy harvesting from the wake since the high streamwise gradient and velocity fluctuations renders increased shear and dilatational turbulent production suitable for energy extraction.

The relationship between the aerodynamic efficiency and the stratified far wake of a turbulent generator was initially explored in Gunasekaran and Altman [1] where the authors showed that the stratified far wake is not just unique to turbulent generators but also that the wing performance characteristics are preserved in the far wake. Understanding this perseverance of performance characteristics requires detailed investigations of near wake turbulent flow properties and their relationship to the aerodynamic efficiency. Hence the near wake of an SD7003 wall-to-wall model was chosen to be interrogated in this study.

Evidence from literature shows that the near wake turbulent properties are dependent on the angle of attack of the wing, the freestream Reynolds number, trailing edge geometry and the wing geometry (camber, thickness, etc.). Each of these variables play a significant role in the symmetry, growth and evolution of spatial and temporal turbulent scales in the near wake. Given that these same variables heavily influence the maximum aerodynamic efficiency location, a possible correlation could exist. Even though the answers to the above posed questions are not explicitly addressed in the literature, a few key findings can be extracted and will be described below.

A. Influence of Reynolds Number in Near Wake

Inviscid theoretical aerodynamics predicts lift coefficients to be independent of Reynolds number effects. However, at low Reynolds number (10,000 – 200,000), viscous effects have a strong influence such that theoretical inviscid models are no longer applicable. Coincidentally, it is at these Reynolds numbers, most airfoils produce tonal and broadband noise [2] from the standpoint of acoustics. At these Reynolds number the dominant viscous effect often makes application of the Kutta condition invalid and flow separation is initiated long before the trailing edge. Mechanisms such as laminar separation bubbles (Rodriguez et. al. [3]), Kelvin-Helmholtz instabilities (Clinton and Browand [4]) and Tollmein and Schlichting (T-S) waves (Baines et al., [5]) play a dominant role in determining the onset of flow separation and near wake turbulent properties.

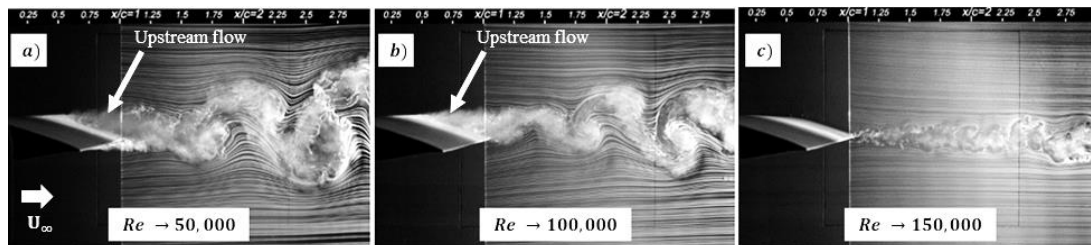


Figure 1 Shedding of vortices from the trailing edge of NACA 0025 airfoil at different Reynolds numbers [6].

Yarusevych, et. al. [6] investigated the changes in near-wake vortex shedding and the corresponding coherent structures with respect to Reynolds number for a NACA 0025 airfoil experimentally. Yarusevych, et. al. found that Reynolds number has an obvious and clear impact on the near-wake coherent structures. Smoke visualizations performed on a NACA 0025 airfoil showed interesting results at three different Reynolds numbers (Figure 1). At

Reynolds numbers of 50,000 and 100,000, clear vortex shedding was observed (Figure 1a and 1b) in the near wake of the airfoil with larger coherent structure and length scales. The smoke visualization shows upstream propagation of flow (since the smoke wire is placed downstream) which indicates that vortex shedding could be responsible for boundary layer flow separation. However, at a Reynolds number of 150,000, the large-scale structures disappeared, and no upstream propagation of flow was observed. Increase in Reynolds number is seen to reduce the coherent structures length scales. Through further analysis, Yarusevych identified a linear dependency between the wake vortex shedding frequency and Reynolds number. However, it is not clear whether this relationship holds for multiple angles of attack or if it changes in maximum aerodynamic efficient conditions. Given that these results were generated by an airfoil of very high thickness, it is suspected that they may not translate to airfoils considered more practical for these Reynolds numbers.

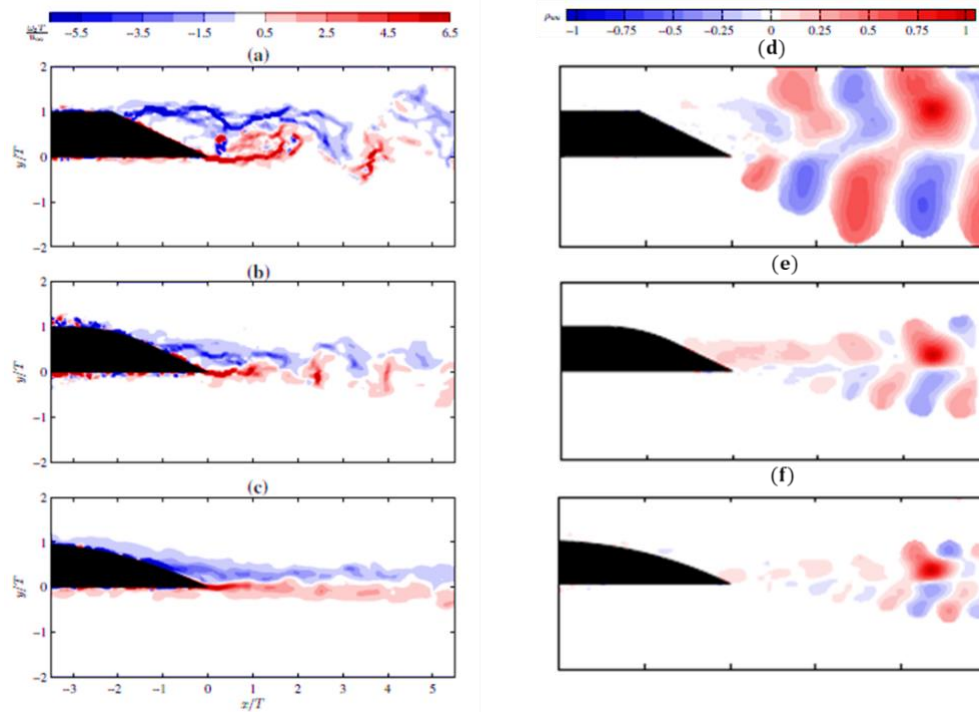


Figure 2 Instantaneous vorticity contours from different trailing edge geometries (a, b, and c). Two-point cross correlation of streamwise velocity in the wake of different trailing edge geometries (d, e and f). Coherent structures can be seen in all the trailing edge geometries [8].

B. Influence of Trailing Edge Geometry

Along with Reynolds number, trailing edge geometry has also been found to change the near wake characteristics as shown by Blake [7] and Guan, et. al [8]. Historically, multiple trailing edge shapes were explored to determine the acoustic tones and radiated sound power as trailing edge noise is one of the main sources of sound in flow over airfoils at low Reynolds numbers. Guan et. al [8] found significant change in near wake coherent structures between sharp and smooth trailing edges as observed in the instantaneous vorticity and streamwise two-point correlation contours (Figure 2). The contours show that a relatively high radius of curvature (Figs. 2c and 2f) causes the flow to remain attached much longer than a sharper trailing edge (Figs. 2a and 2d). The latter of which causes separation to occur much earlier. Although coherent structures are present for all trailing edge geometries, there is a clear reduction in length scale seen with increased radius of curvature.

C. Influence of Angle of Attack

The influence of angle of attack on the near wake was explored through several experimental and computational investigations. The DNS simulation of a NACA 0012 airfoil at different angles of attack at a Reynolds number of 50,000 by Sandberg et. al [9] revealed that at zero incidence angle, the flow is symmetric on the top and bottom surface except for the small amplitude T-S waves which indicate the onset of turbulent flow transition. It should be noted that

the first sign of instability occurs in the near wake region as the free shear layer responds to background perturbation. Shan et. al [10] described that the disturbances in the near wake propagate upstream in the form of acoustic waves thereby perturbing the upper surface shear layer. At higher angles of attack, a laminar separation bubble was formed and vortex shedding occurred on the suction side of the airfoil as shown in Figure 3b. These vortices produce pressure disturbances that convect over the surface of the airfoil and pass over the trailing edge. The streamwise growth of the pressure disturbances results in oscillations of the separated shear layer and eventually leads to vortex shedding. With the increase in angle of attack, Sandberg et. al [9] noted that the formation of separation bubbles and the onset of vortex shedding happens considerably farther upstream when compared to lower angle of attack cases.

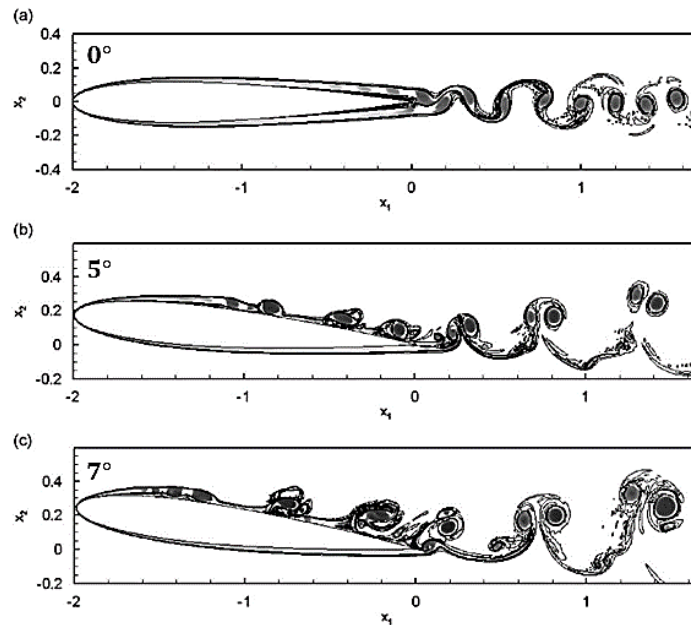


Figure 3 Instantaneous contour lines of the spanwise vorticity component for NACA 0012 at $\alpha = 0^\circ$ (a), $\alpha = 5^\circ$ (b) and $\alpha = 7^\circ$ (c) at $Re = 50,000$. [9]

D. Influence of Airfoil Geometry

It is known that an adverse pressure gradient due to curvature contributes to a larger wake spreading rate and Reynolds stress. Experiments done by Hah and Lakshminarayana [11], Nakayama [12] and Weygandt and Mehta [13] confirm that even mild changes in curvature have significant effect on the mean and turbulent quantities in the wake. Zhang et. al [14] investigated the geometry effects on flow separation and near wake properties of NACA 0012 and NACA 4412 airfoils at a Reynolds number of 10,000. As expected, Zhang et. al showed conclusive evidence that increase in curvature creates greater adverse pressure gradient and leads to vortex shedding. The off-diagonal Reynolds stress contour indicated that the maximum Reynolds stress was achieved along the edges of the displacement thickness of the airfoil. These simulations were performed at 10-degree angle of attack for both wings, however, beyond the maximum aerodynamic efficiency condition of NACA 4412.

E. Influence of Wake Symmetry and Connection with Far Wake

As mentioned earlier, Gunasekaran and Altman [1] demonstrated that the far wake turbulent characteristics are not only unique, but the wing performance information is also preserved in it. However, a strong asymmetry was observed in the wake even after 10 chord lengths downstream of the wing. Similar properties were observed both in the upper and lower surface shear layer. Better insight into this relationship and the connection between the upper and lower surface shear layer can be obtained through further investigation into the near wake and by determining the influence of geometry, angle of attack and Reynolds number on the wake symmetry. Studies have shown that a symmetric and an asymmetric wake have different behaviors. Considering the classical case of a flat plate, Thomas and Liu [15] experimentally investigated asymmetric and symmetric wakes. It was found that these types of wakes had inherent differences from each other. Asymmetric wakes were found to spread faster than symmetric ones under similar conditions, suggesting that asymmetry can play a role in how the wake develops. It was also shown that an asymmetric

wake's mean velocity profile (momentum deficit) tends to approach a more symmetric distribution as the wake develops downstream. This mean velocity profile behavior was also seen in the turbulent kinetic energy. In other words, initially asymmetric turbulent kinetic energy (TKE) profiles were shown to become more symmetric with downstream distance. Turbulent kinetic energy is dependent on both mean and fluctuating quantities. This trend in mean velocity and TKE may indicate the fluctuating quantities have a similar tendency to approach symmetry with downstream distance as well. However, the results from Gunasekaran and Altman [1] showed that the asymmetry persists even after 10 chord lengths downstream of the wing. This deviation could be due to the multitude of variables which affect the structure, mean and fluctuating quantities in the wake.

The literature review shows that determining a relationship between the aerodynamic efficiency and wake characteristics as a function of all the variables: angle of attack, geometry, trailing edge extensions and Reynolds numbers can open a plethora of potential experimental investigations. To start with the parametric analysis, the research presented in this paper is intended to isolate the effect of angle of attack on the near wake properties while maintaining a constant Reynolds number.

The primary goal of this investigation is to provide insight into the relationship between the mean and fluctuating flow quantities in the near wake and the aerodynamic efficiency as a function of angle of attack of SD7003 airfoil along with wake symmetry investigations.

II. Experimental Test Setup

A. Horizontal Free-Surface Water Tunnel

Time Resolved Particle Image Velocimetry (TR-PIV) experiments were conducted in the near wake of a wall-to-wall SD7003 wing model at the Air Force Research Lab's (AFRL's) Horizontal Free-surface Water Tunnel (HFWT). The geometry and flow characteristics of the SD7003 wing and HFWT are shown in Table 1.

Table 1. Specifications of the tested SD7003 wall-to-wall model

Chord	152.4 mm
Semi-span	228.6 mm
Max t/c Ratio	0.085
Freestream Velocity	0.4 m/s
Chord based Reynolds Number	~68,000

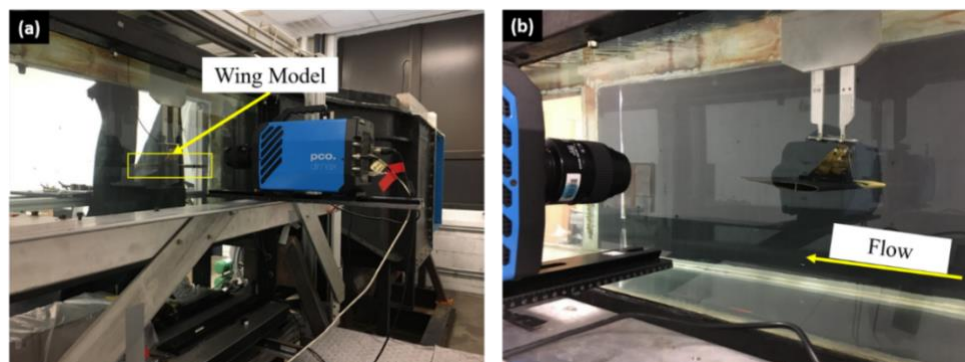


Figure 4 Two views of the HFWT experimental setup oriented (a) upstream and (b) downstream of the inlet.

The HFWT has a 4:1 contraction ratio with an 18 in. wide (46 cm.) x 24 in. (61 cm.) tall test section with a freestream velocity range of 3 – 105 cm/s. The tunnel is fitted with a three degree of freedom motion rig (Figure 4) consisting of a triplet of H2W linear motors, driven by AMC DigiFlex servo-drives controlled by a Galil DMC 4040 4-channel card, with user-selected proportional/integral/derivative (PID) constants for each channel. The model's pitch is controlled via two motors mounted vertically on a plate above the tunnel test section. Each motor actuates a vertical "plunge rod", which connects via a bushing to a coupler piece holding the force balance, and ultimately the test article. For these tests the upstream plunge rod is constrained to move purely vertically by a nylon track, whereas the downstream plunge rod can pivot in the test section vertically about a plane of symmetry.

The wing model was inverted with the suction side facing the floor of the test section (previous Figure 4b.) to avoid mount-interference effects on the upper surface. The 0° angle of attack baseline was set by digitally measuring the angle of attack of the wing relative to a horizontal line. This resulted in an uncertainty in angle of attack of ± 0.1 degree.

B. PIV Setup

A PCO DiMax camera with a 2000 x 2000 pixel array, was mounted on a one-degree of freedom rail. A 105 mm Micro Nikon lens was used and the camera was operated via the PCO CamWare software. The size of the field of view behind the trailing edge of the wing was square at 75 mm x 75 mm ($0.5 \bar{c} \times 0.5 \bar{c}$) resulting in a spatial resolution of 26 pixels/mm. Titanium dioxide (TiO_2) particles were used to seed the HFWT, which have a mean scattering diameter of 10 microns. A ~ 50 mJ/pulse frequency doubled Nd:YLF high speed MetroLaser was used to illuminate the seed particles.

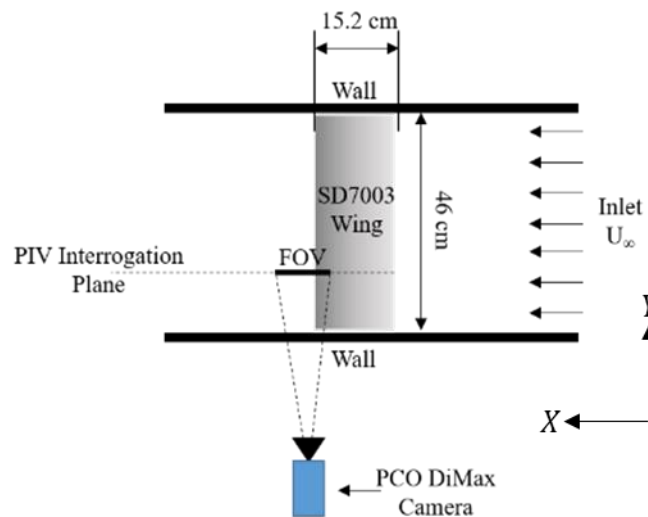


Figure 5 Schematic of the TR-PIV HFWT test setup with mid-semi-span interrogation region laser alignment.

The laser passed through a converging and diverging lens to open the beam into a sheet, with a mirror used to reorient the laser sheet from horizontal to vertical. In order to investigate the behavior of the upper and lower shear layers in the near wake of the wing, the trailing edge of the wing was included in the PIV interrogation window. TR-PIV was performed at the mid-semi-span (4.5 in. inboard along the spanwise direction), as illustrated in the schematic shown in Figure 5, at the angles of attack shown in Table 2. The time delay, Δt , was tuned to ensure a mean 8 – 10-pixel particle displacement at each angle of attack in the wake. This resulted in a Δt of 0.844 milliseconds (1.18 kHz framerate) until 3° angle of attack and was increased to 0.926 milliseconds (1.08 kHz framerate) for angles of attack greater than 4° .

Table 2. Test matrix for PIV on the SD7003 wall-to-wall model

PIV Interrogation Plane	Mid-semi-span (4.5 in. inboard)	
Angles of Attack (deg.)	-2, 0, 1, 2, 3	4, 5, 6, 7, 8
Δt (ms)	0.844	0.926
Framerate (kHz)	1.18	1.08

For each angle of attack case, a minimum of 1000 image-pairs were analyzed and processed using Innovative Scientific Solutions, Inc.'s (ISSI) DPIVB software. Cross-correlation was performed, for all cases, with a two-pass interrogation region size of 64 x 64 pixels on the first pass and 32 x 32 pixels on the second. The overlap between the interrogation windows in each iteration was 50%. An in-house MATLAB script was used for further analysis and computation of the flow quantities. Additionally, a sensitivity analysis was performed on the PIV data in order to

account for radial distortion effects and the water tunnel's relatively high (compared to most airplane wind tunnels) freestream turbulence intensity levels.

III. TR-PIV Results

The following section discusses the behavior of the near wake flow properties as analyzed using the PIV setup described in the previous section. Discussion will focus on the momentum deficit, instantaneous vorticity, two-point correlation in the streamwise and transverse directions, the energy density spectrum, root-mean-square velocities, and Reynolds shear stress.

Each property was analyzed at three downstream distances by taking vertical profile slices. The location of each position is shown in Figure 6 and covers, in total, approximately $\frac{1}{2}$ chord length from the SD7003 trailing edge. Eleven columns total were averaged at each y-coordinate in the vertical profile cut: 5 columns on either side plus the slice-location itself. The y-position in all profile plots was normalized by the wake half-width.

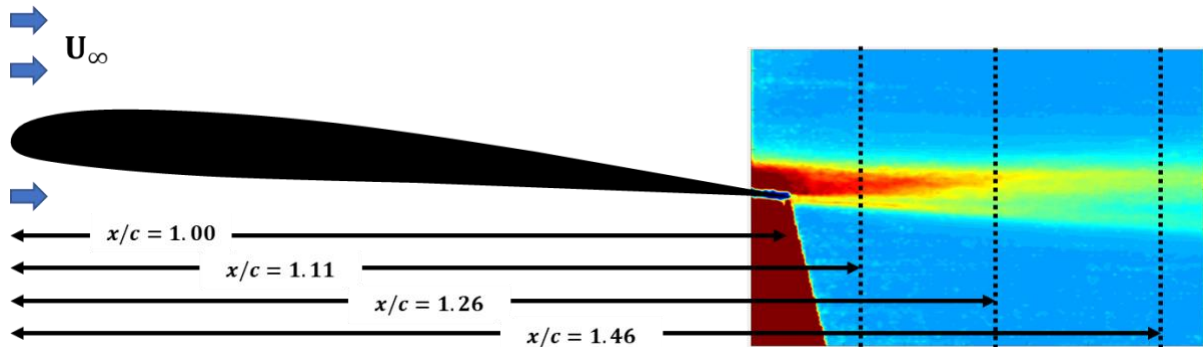


Figure 6 Schematic shown three locations of vertical profile cuts used to analyze each flow property.

A. Momentum Deficit

The normalized plots of the momentum deficit distribution across all angles of attack at each of the three downstream locations is shown in Figure 7. As expected, the momentum deficit, in general, increases with increase in angle of attack. However, there is a notable difference in the trend of momentum deficit profiles with angle of attack and with the increase in the downstream distance. Closer to the trailing edge, the near wake velocity distribution has greater asymmetry at angles of attack greater than 2° . As downstream distance increases, the distribution approaches greater symmetry with comparatively lower skewness and kurtosis. This result is in line with Thomas and Liu's [1] experimental results for an asymmetric wake behind a flat plate.

Examining the three plots in Figure 7, the momentum deficit profiles can be separated into three different groupings which are represented in three separate colors. The groupings are especially prevalent closer to the trailing edge, although the phenomenon remains with increasing distance downstream as well. Three different groups were observed as described in Table 3.

Table 1 Three different groups observed in momentum deficit profiles

Group	AOA (Degrees)	Color Representation in Figure 5
1	-2, 0, 1	Blue
2	2, 3, 4, 5	Green
3	6, 7, 8	Red

Group 1 which contains angles of attack from -2° to 1° is represented in blue. Group 2 containing angles of attack from 2° to 5° is represented in green and Group 3 containing angles of attack from 6° to 8° is represented in red. A notable difference in these groupings, particularly at the trailing edge (Figure 7a) is the change in the skewness and kurtosis of the Gaussian distribution from Group 1 to Group 2. The kurtosis drastically decreases, and skewness

increases dramatically as well. These changes are indicators of greater wake asymmetry. The change between Group 1 and Group 2 occurs from 1° to 2° angle of attack, which is astonishing. The nature of the wake changes significantly within a 1° angle of attack increment. At higher downstream distances, the differences between the momentum deficit profiles decrease, but the angle of attack grouping listed in Table 3 persists. At a downstream location of $x/c = 1.46$, the momentum deficit profiles show similar trends with changes in angle of attack. However, the profiles at lower angles of attack show symmetry whereas at higher angles of attack, the profiles are asymmetric. In addition to the change in overall shape of the normalized momentum deficit profile, there is a greater decrement in peak velocity (observed by the peak U_x velocity) from 6° to 8° angles of attack. This decrement can be clearly observed in Figure 7b and Figure 7c. It is this sudden reduction in momentum deficit which motivated the formation of a separate group (Group 3) for angles of attack greater than 6° .

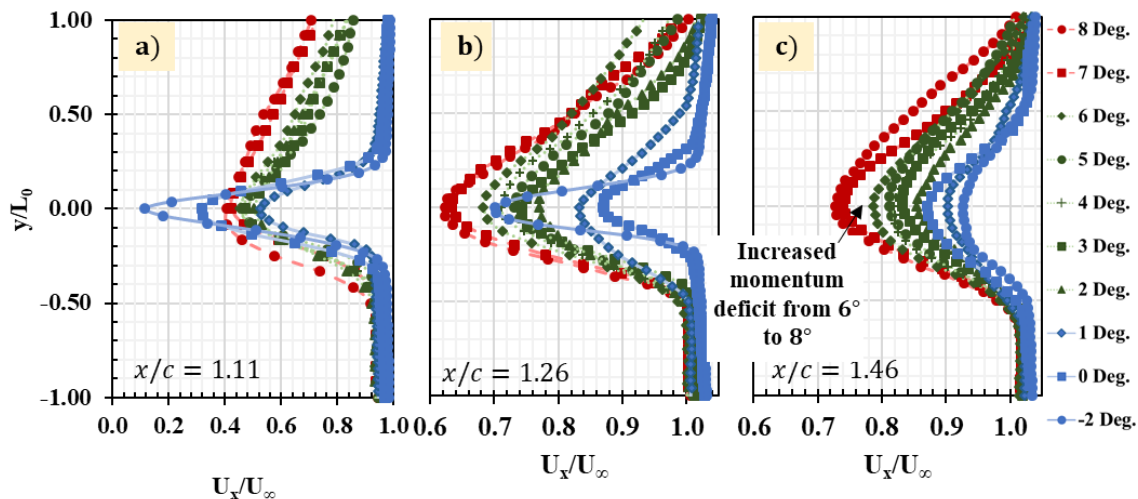


Figure 7 Normalized momentum deficit for all angles of attack at Locations (a) 1 (trailing edge), (b) 2 ($\frac{1}{4}$ chord downstream), (c) 3 ($\frac{1}{2}$ chord downstream).

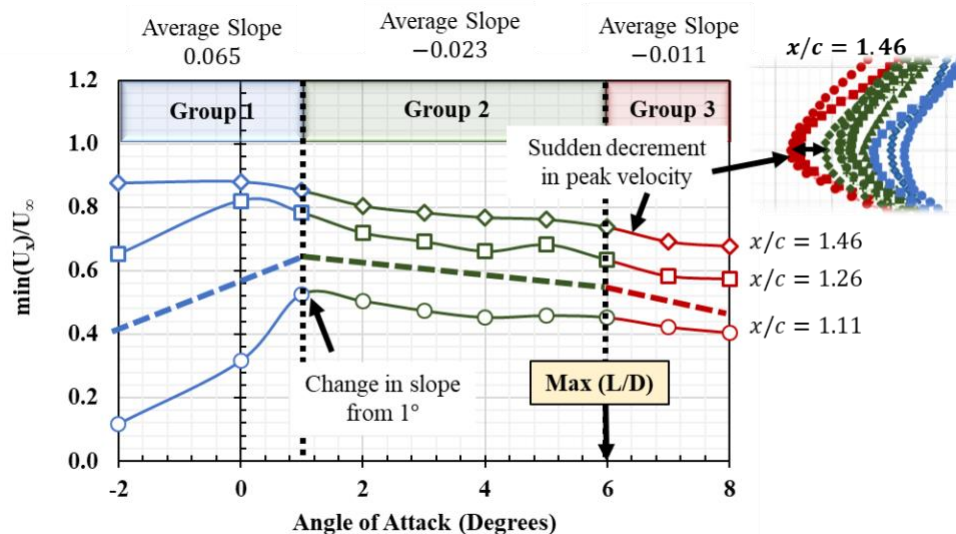


Figure 8 Plot of normalized peak momentum deficit, $U(y)/U_\infty$, at all angles of attack for all profile locations.

The distinction between the different groups can be viewed differently when plotting the normalized peak momentum deficit values, as shown in Figure 8. With angles of attack less than 1° , the slope of the peak momentum deficit values has higher magnitude. At the angles of attack between 2° and 5° , the peak momentum deficit is decreasing consistently. At an angle of attack beyond 5° , the slope decreases even further. However, the change in the slope between Group 2 and Group 3 is not as high as the change in slope from Group 1 and Group 2.

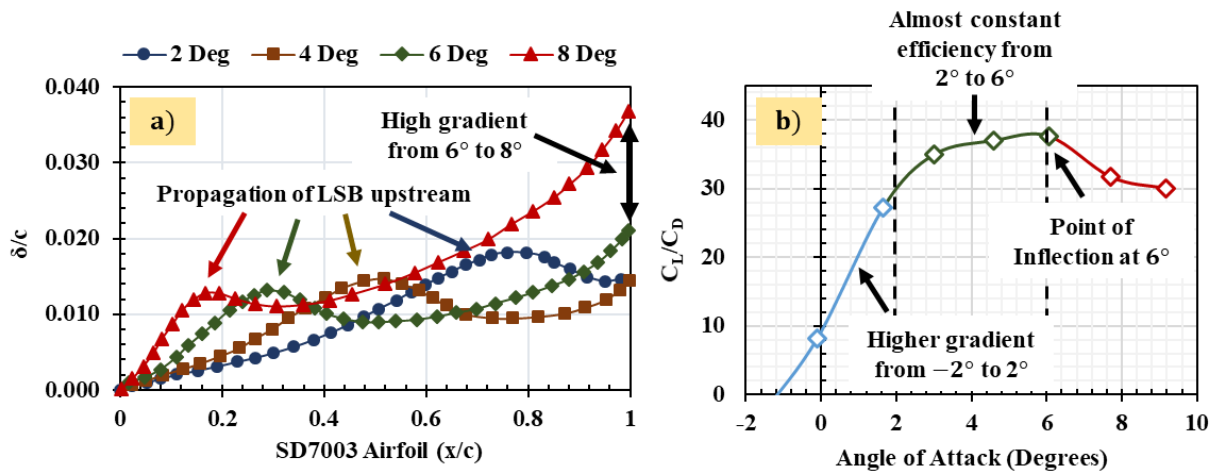


Figure 9 a) Variation of boundary layer thickness as a function of angle of attack. b) Variation of C_L/C_D with increase in angle of attack (Selig et. al [16])

To get insight into the different groupings, the displacement thickness on the upper surface boundary layer was determined using viscous XFLR5 simulation [17] at a Reynolds number of 60,000 (Figure 9a). At a 2° angle of attack, the formation of the laminar separation bubble (LSB) can be seen as increased displacement thickness near the trailing edge of the airfoil. At angles of attack lower than 2° , no such LSB was seen. The changes in momentum deficit profiles from Group 1 to Group 2 could indicate the onset of the formation of the LSB. Coincidentally, different groupings can also be seen in the aerodynamic efficiency of the SD7003 airfoil obtained from Selig et. al [16] for a Reynolds number of 60,000. There is a steep rise in the aerodynamic efficiency up to a 3° angle of attack shown in Figure 9b. After 2° angle of attack, the aerodynamic efficiency plateaus. At angles of attack between 2° and 6° , the LSB propagates upward towards the leading edge of the airfoil as annotated in Figure 9a. At these angles of attack, the displacement thickness at the trailing edge of the airfoil remains constant. However, at angles of attack greater than 6° , the displacement thickness at the trailing edge increases significantly. Coincidentally, a steep decrement is also seen in the aerodynamic efficiency from 6° to 8° angle of attack shown in Figure 9b. In order to understand the changes in the flow field in the different groupings, the instantaneous vorticity and 2-point correlation are discussed below.

B. Evolution of the Instantaneous Vorticity Field

A fourth-order finite difference scheme was used to calculate the vorticity. This scheme was selected because of its success with experimental data due to less smoothing of gradients when compared to a second-order least square method, and due to the scheme producing a small truncation error [18], [19]. Vorticity, using a fourth-order finite difference, is defined as:

$$\omega_{i,j} = \frac{1}{12\Delta X}(-v_{i+2,j} + 8v_{i+1,j} - 8v_{i-1,j} + v_{i-2,j}) - \frac{1}{12\Delta Y}(-u_{i,j+2} + 8u_{i,j+1} - 8u_{i,j-1} + u_{i,j-2}) \quad (1)$$

Where u , v , x , and y refer to the respective velocities and directions. Additionally, i and j refer to coordinates representing some spatial location within the 2D plane used to calculate vorticity. To study the temporal evolution, vorticity was calculated individually for each image-pair's vector data. Instantaneous vorticity contours are shown in Figure 10 for multiple angles of attack. Only one angle of attack in each group is shown in Figure 10 for comparison. Vortex shedding is most clear at the 0° angle of attack. It should also be noted that the vortex shedding is seen in -2° and 1° angles of attack as well. In the -2° case, the onset of shedding vortices occurs downstream of the trailing edge, whereas 0° angle of attack indicates a strong shedding pattern close to the trailing edge. Comparing the displacement thickness variation, it appears that the vortex shedding is observed before the formation of the LSB. And the aerodynamic efficiency at these conditions is comparatively lower as seen in Figure 9b.

However, with increasing angle of attack, the vortex shedding pattern disappears as shown in Figure 10b and the LSB starts to appear over the upper surface of the airfoil. Within 1° angle of attack, the nature of the near wake and the boundary layer thickness change due to the formation of the LSB near the trailing edge. Beyond 1° angle of attack, the strength and clarity of the vortex shedding diminishes greatly. An increase in the boundary layer thickness is seen with increasing angle of attack, evident through the thickening of the vorticity band on the upper surface (blue).

At higher angles of attack, 3° and up, there remains some semblance of a pattern, however, it does not appear to be periodic vortex shedding. Analyzing the coherent structures provided more insight into this pattern and will be discussed in the next section. Trends were also found in wake thickness variation as a function of angle of attack. The wake thickness is quantified by determining the 99% U_∞ threshold. At lower angles of attack, the vortex shedding causes the wake to be thicker. This thickness increases as a function of downstream distance. Additionally, the wake thickness decreases with increase in angle of attack. At angles of attack greater than 6° , the wake thickness increases along with a corresponding increase in vorticity magnitude. The reduction in the aerodynamic efficiency at angles of attack higher than 6° could be due to the onset of flow separation at the trailing edge as indicated by a higher displacement thickness as shown in Figure 9a.

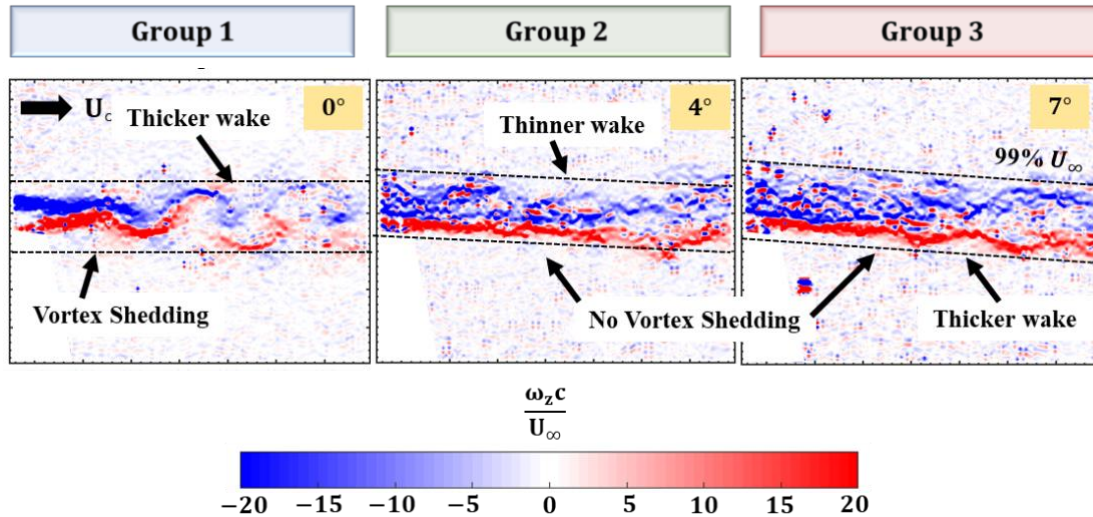


Figure 10 Instantaneous vorticity at 0° , 4° and 7° angles of attack representing each groupings.

C. Streamwise and Transverse Two-Point Correlation

A two-point correlation, denoted by ρ , allows for investigation into the evolution of the wake downstream of the wing with respect to a reference point. Primarily, two-point correlation will offer insight into the growth and formation of coherent structures in the near wake and length scale variation under different angle of attack conditions. It is possible to correlate all points in a given flow region either spatially or temporally. That is, the evolution of the wake may be examined with respect to space or time. In general, the two-point correlation is defined in [19] as:

$$\rho_{u_i, u_j}(x_1, x_2, \tau) = \frac{\overline{u'_i(x_1, t) u'_j(x_2, t + \tau)}}{\sqrt{\overline{u'^2_i(x_1)}} \sqrt{\overline{u'^2_j(x_2)}}} \quad (2)$$

where u' refers to the fluctuating velocity components in either the “ith” or “jth” direction (i.e., streamwise or transverse); t refers to the time and τ is some time-step. In this study, τ is assumed to be zero such that a spatial, rather than temporal correlation was performed. The variables x_1 and x_2 refer to the reference point and, iteratively, all other points in the flow, respectively.

Figures 11 details the two-point correlation performed in the SD7003 wake in both the streamwise, ρ_{uu} , and transverse, ρ_{vv} , directions respectively, for angles of attack representative of each grouping (Group 1, 2, and 3). It should be noted that Group 1 has a larger contour range (-0.5, 0.5) than those angles of attack in Groups 2 and 3 (-0.25, 0.25). This was necessary to capture the correlation strength which likely results from the periodic vortex shedding seen in that angle of attack range. The reference point is concentrated in the upper surface boundary layer. The coordinates themselves vary with angle of attack due to downwash at higher angles. Reference points are denoted by a star in each contour.

These contours provide insight into the turbulent length scales as well as the relative vortex shedding frequency. As a result of strong vortex shedding at 0° , as seen in the instantaneous vorticity discussion, a high correlation strength is present relative to the other angle of attack groupings. This clear periodic vortex shedding, based on the analysis so

far, appears to be indicative of Group 1, or the lower angles of attack (-2° to 1°), with periodic shedding no longer present in Group 2 or 3. Similar behavior is shown in the streamwise two-point correlation as well: with a clear alternating pattern present in Group 1, which is absent in Groups 2 or 3. Additionally, the 8° angle of attack case being representative of Group 3 has a lower correlation strength due to the onset of stall. In general, the length scales can be seen to decrease with increasing angle of attack as well.

It has been shown by Yarusevych, et. al. [6] that development of coherent structures in an airfoil wake may not be apparent when examining the U-velocity fluctuating component but may appear in the transverse velocity (V-velocity) fluctuating component. This suggests that, although the ρ_{uu} correlation may not indicate clear alternating patterns, the ρ_{vv} correlation strength could offer additional insight into the wake characteristics.

At 0° angle of attack, the transverse two-point correlation has an even stronger magnitude when compared to the streamwise correlation. This is another indication of the vortex shedding pattern seen in the instantaneous vorticity contours. At 5° , it is possible to see a pattern in the transverse correlation. This could suggest, as described by Yarusevych, that some amount of vortex shedding is present, but only evidenced by the V-component of velocity. However, no shedding pattern was found in the instantaneous vorticity contours at 5° angle of attack. As with the streamwise, the transverse correlations are relatively weak at 8° . This could be an indication of stall onset. Figure 11 shows a further decrease in length scale with increasing angle of attack. Based on both streamwise and transverse two-point correlations, it appears that larger length scales are characteristic of Group 1 (-2° , 0° , and 1°). Following a similar trend as the momentum deficit, there appears to be a magnitude change from Group 2 (2° to 5°) to Group 3 (6° to 8°). However, while momentum deficit increased in Group 3, the two-point correlation suggests a *decrease* in both correlation strength and length scale moving from Group 2 to Group 3 keeping in trend with the viscous cascade.

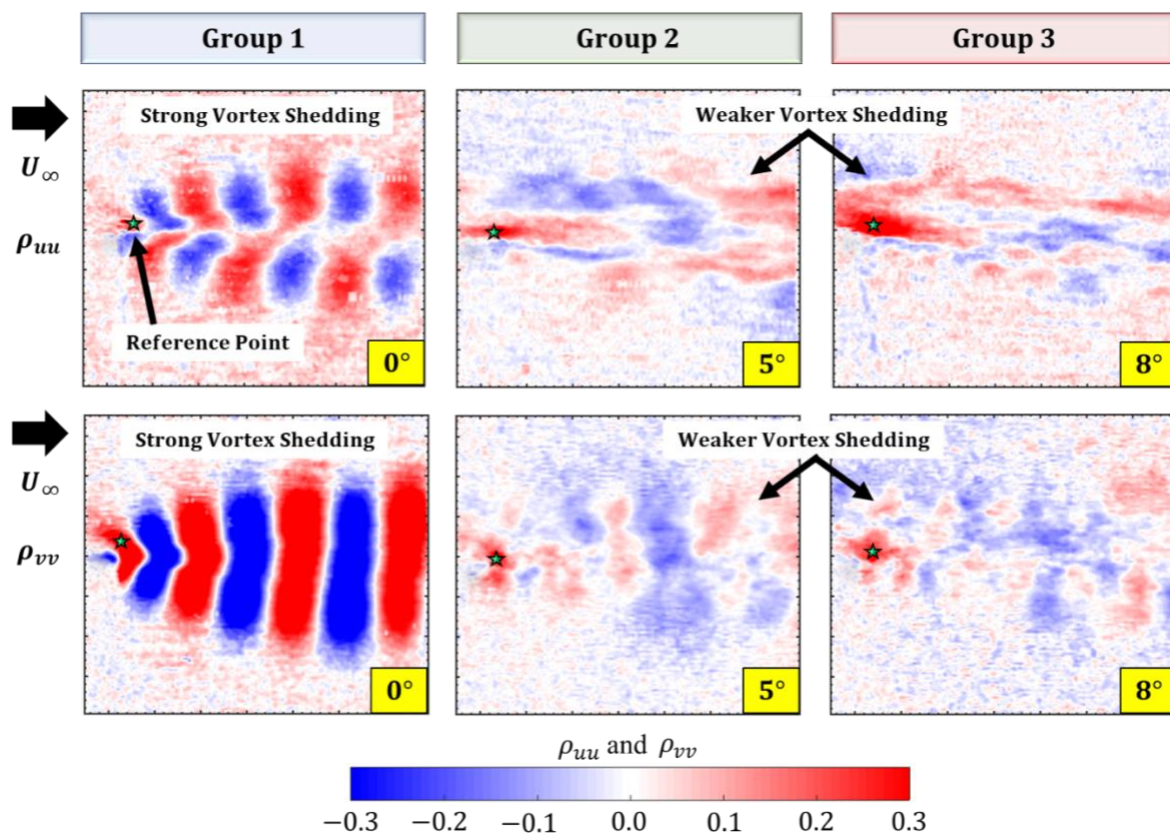


Figure 11 Two-point correlation of streamwise and transverse velocity components in the near wake.

C. Energy Density Spectrum

A Fast Fourier Transform (FFT) was performed on the streamwise velocity fluctuations in order to analyze the energy density spectrum. Reference points used were identical to those chosen for the two-point correlation, discussed in the previous section. Figure 12 contains plots of the kinetic energy in the frequency, rather than temporal, domain.

These plots are for angles of attack representative of each grouping: 0°, 5°, and 8°. A line of best fit was plotted alongside the energy spectrum.

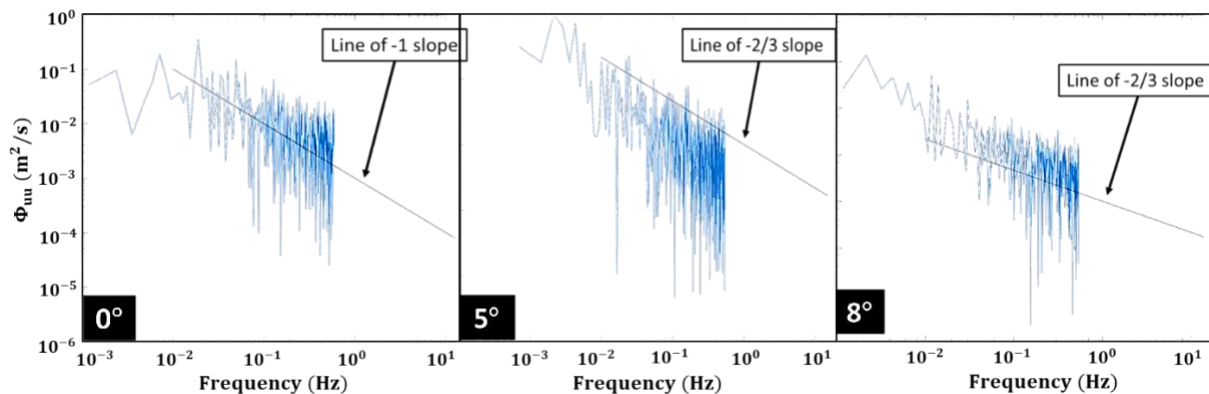


Figure 12 Energy spectral densities in the near wake on a representative angle of attack from each group.

As expected, Figure 12 indicates a lack of agreement with Kolmogorov's $-5/3^{\text{rd}}$ Law, which assumes homogeneous turbulence. With homogeneous turbulence, the properties do not change from point to point. However, this is not the case for our interrogation region, as the properties do change. It appears that the slope of the energy spectrum changes from Group 1 to Groups 2 and 3, with Group 1 having a faster energy decay rate when compared with Groups 2 and 3. This variation in slope provides further evidence for a differing wake signature between the lower angles of attack, with vortex shedding, and those higher angles of attack, for which shedding is not present.

IV. Fluctuating Flow Quantities in the Near Wake

A. Streamwise Fluctuating Velocity Component

Figure 13 displays the contours of normalized U_{RMS} at each angle of attack tested in this study. The Group 1 angle of attack case can be seen to have a relatively symmetric distribution. This symmetry quickly breaks down as the angle of attack is increased, resulting in an asymmetric distribution. As with the mean quantities, the wake appears to approach a symmetric distribution, in terms of U_{RMS} magnitude, with increasing downstream distance.

Additionally, the contours below show a clear decrease in normalized U_{RMS} magnitude after 0° angle of attack. This trend is seen in the momentum deficit as well, suggesting that this behavior may be indicative of Group 2 (2° to 5° angle of attack). A decrease in U_{RMS} magnitude around the maximum aerodynamic efficiency angle is visible as well when transitioning from Group 1 to Group 2. As with the mean flow quantities examined, Group 3 is characterized by an increase in magnitude beyond the maximum aerodynamic efficiency condition. The increase in the streamwise fluctuating nature of the wake from 5° to 8° is likely due to onset of flow separation, as the airfoil is stepped closer to its stall angle.

In order to more closely examine the trend in U_{RMS} , vertical profile slices were taken at the trailing edge, $1/4$ chord downstream, and $1/2$ chord downstream and are shown in Figure 14. At 0 degrees angle of attack, particularly at the trailing edge (Figure 14a), the U_{RMS} magnitude is large compared to the other angles of attack. This increased U_{RMS} magnitude is the result of the periodic vortex shedding seen in the instantaneous vorticity contours. In fact, at lower angles of attack (Group 1) an increased U_{RMS} magnitude, relative to the higher α 's, is noted. This increased RMS seems to be an effect of the periodic vortex shedding patterns seen at all angles in Group 1.

As with the momentum deficit profiles, distinct zones (groupings of angles of attack with similar slope) are found in the normalized U_{RMS} profiles. Again, Group 1 consists of the lowest angles of attack considered here: -2°, 0°, and 1°, while Groups 2 and 3 consist of 2°–5° and 6°–8° angles of attack, respectively. These similar groupings would suggest a potential correlation between the mean and fluctuating components of the SD7003 wake.

To further investigate the streamwise fluctuating velocity component, plots of the upper surface normalized peak U_{RMS} values were generated at all three downstream locations. These plots can be found in Figure 15. As seen in the Figures below, the 0° case shows a marked increase in U_{RMS} , which is expected due to the pattern of periodic vortices being shed in the wake from the SD7003's trailing edge.

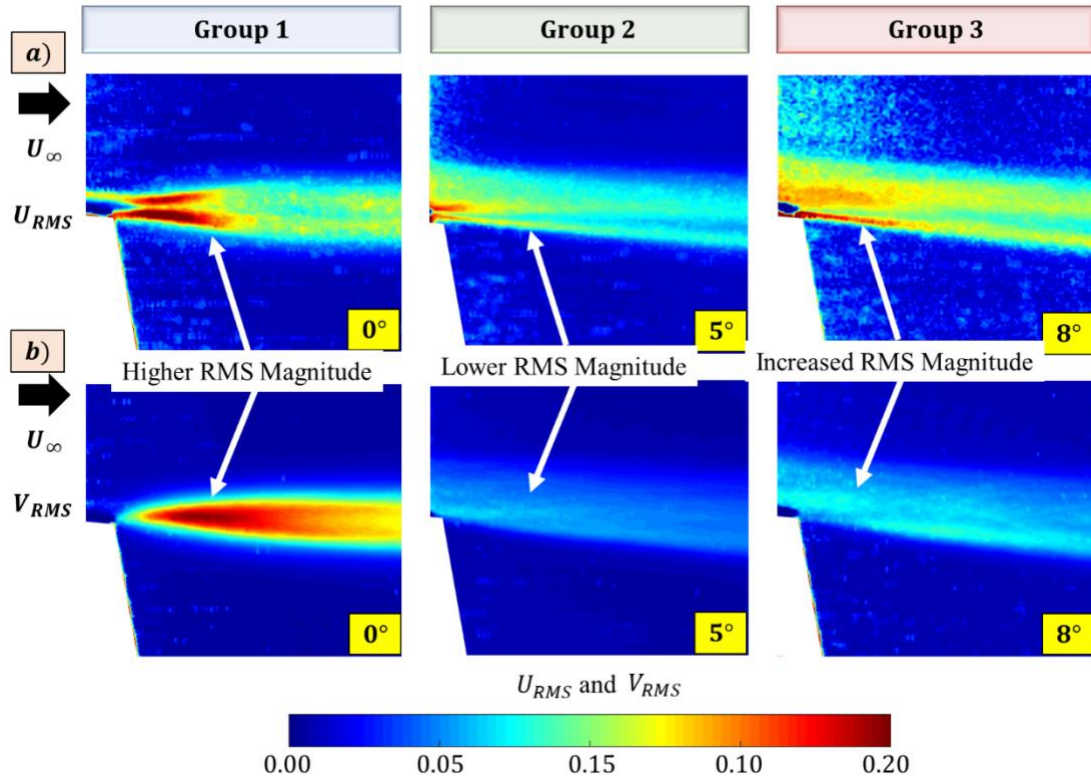


Figure 13 URMS and VRMS contours for different angles of attack.

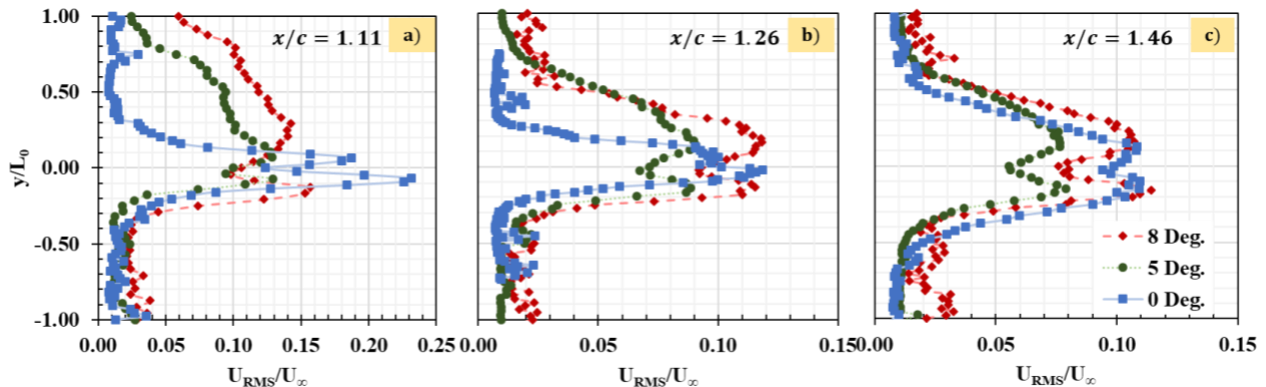


Figure 14 Comparison of normalized U_{RMS} for representative angles of attack at Locations (a) 1 (trailing edge), (b) 2 ($\frac{1}{4}$ chord downstream), (c) 3 ($\frac{1}{2}$ chord downstream).

Particularly unique about the U_{RMS} trends is the point of inflection which is seen from 6° to 8° angle of attack. This is not something that was seen in the mean quantities; although a change in magnitude was present, a high gradient appears only in the streamwise fluctuating velocity component so far. This high gradient shows a clear distinction between Group 2 and Group 3, which has a transition point around the maximum aerodynamic efficiency condition. However, the difference between Groups 1 and 2 is apparent in both the mean flow quantities and the U_{RMS} . The former grouping has been shown to be indicative of periodic vortex shedding while the latter does not experience this behavior. This would suggest that the transition from Group 1 to Group 2 is evident across both mean and fluctuating flow properties, whereas Groups 2 and 3 can be clearly identified in the fluctuating quantities only.

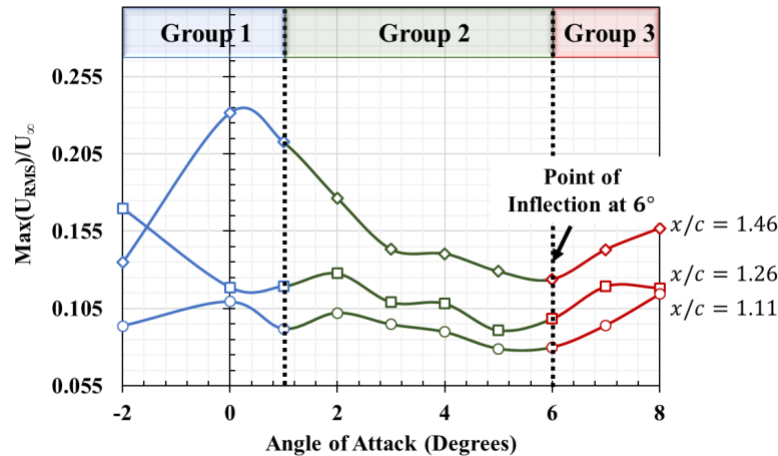


Figure 15 Comparison of normalized U_{RMS} for all angles of attack along the upper surface.

B. Transverse Fluctuating Velocity Component (V_{RMS})

Coupled with velocity fluctuations in the streamwise direction are velocity fluctuations in the transverse, or Y, direction. This component of the wake, V_{RMS} , was studied using a similar methodology as investigation of the streamwise fluctuations from the previous section. Figure 13b contains contours of the normalized V_{RMS} in the SD7003 wake. As expected, the 0° angle of attack contour indicated a marked increase in the transverse fluctuations; as before, this is attributed to the strong periodic vortex shedding present at 0 degrees.

The contours indicate a decrease in normalized V_{RMS} as the angle of attack increases from 0 degrees. From 0° onward, the V_{RMS} magnitude decreases until the maximum aerodynamic efficiency condition. Again, this region approximately comprises Group 2, with the decreased magnitude in the range from 2° to 5° angle of attack. The V_{RMS} magnitude also appears to rise as the angle of attack increases beyond 5° . As with the normalized U_{RMS} , three groupings appear: high magnitude (-2° to 1°) due to the strong vortex shedding; Group 2 shows a decrease in magnitude from 2° to 5° ; and, lastly, Group 3 (6° to 8°) reflects an increase in V_{RMS} magnitude as well.

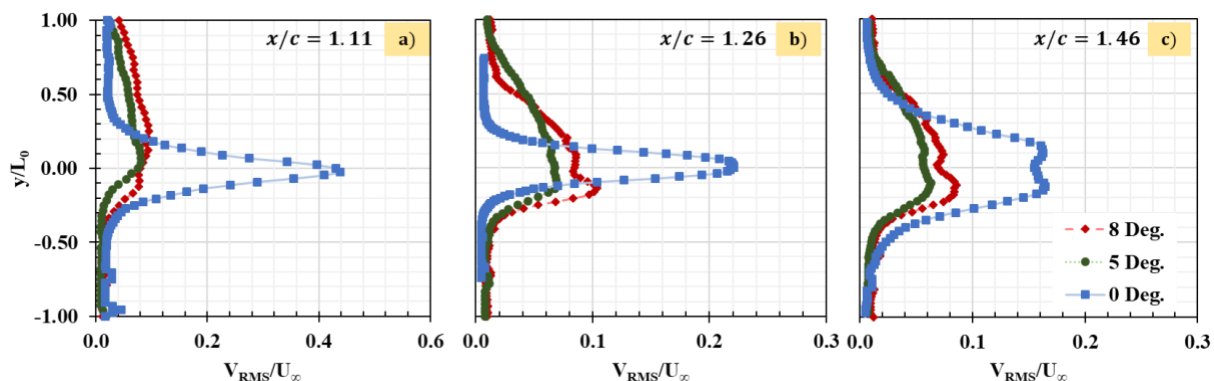


Figure 16 Comparison of normalized V_{RMS} for all angles of attack at Locations (a) 1 (trailing edge), (b) 2 ($\frac{1}{4}$ chord downstream), (c) 3 ($\frac{1}{2}$ chord downstream).

For each angle of attack in question, profile plots at the trailing edge, $\frac{1}{4}$ chord downstream, and $\frac{1}{2}$ chord downstream were generated for the V_{RMS} data. Figure 16 contains these profiles for all angles of attack at each downstream location. Although not as easily discerned as the U_{RMS} and momentum deficit profiles, the V_{RMS} profiles can be separated into zones as well. This is especially visible at the trailing edge (Location 1) where, again, the -2° , 0° , and 1° angle of attack cases demonstrate similar trends, while differing from the other angles of attack in this study (2° to 8°). With V_{RMS} , however, these groupings quickly dissipate with downstream distance. To investigate the normalized V_{RMS} further, a plot at each downstream location of the peak V_{RMS} magnitude can be seen in Figure 17. As

has been discussed, the 0° angle of attack case has clear coherent structures as demonstrated by periodic vortex shedding in the wake. The significantly higher peak V_{RMS} magnitude seen in the Figure below, is likely an effect of the vortex shedding found at lower angle of attack.

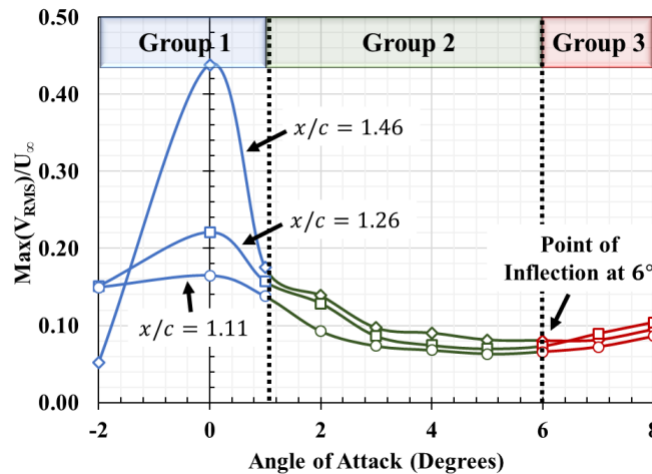


Figure 17 Plot of normalized V_{RMS} at all angles of attack for all profile locations.

The above plot indicates a decrease in peak normalized V_{RMS} from 0° until near the maximum aerodynamic efficiency condition: 5° – 6° angle of attack. This corresponds to similar trends seen for the Group 2 angles of attack in the momentum deficit and U_{RMS} values. Again, a point of inflection is shown from 6° to 8° angle of attack, which corresponds to the increased magnitude that appears to be evidence of the Group 3 angles of attack.

C. Reynolds Shear Stress

The Reynolds shear stress was studied to determine if a similar trend exists with respect to angle of attack. Unlike U_{RMS} and V_{RMS} , the Reynolds shear stress quantifies the overall unsteadiness in the flow while also offering insight into surface pressure fluctuations.

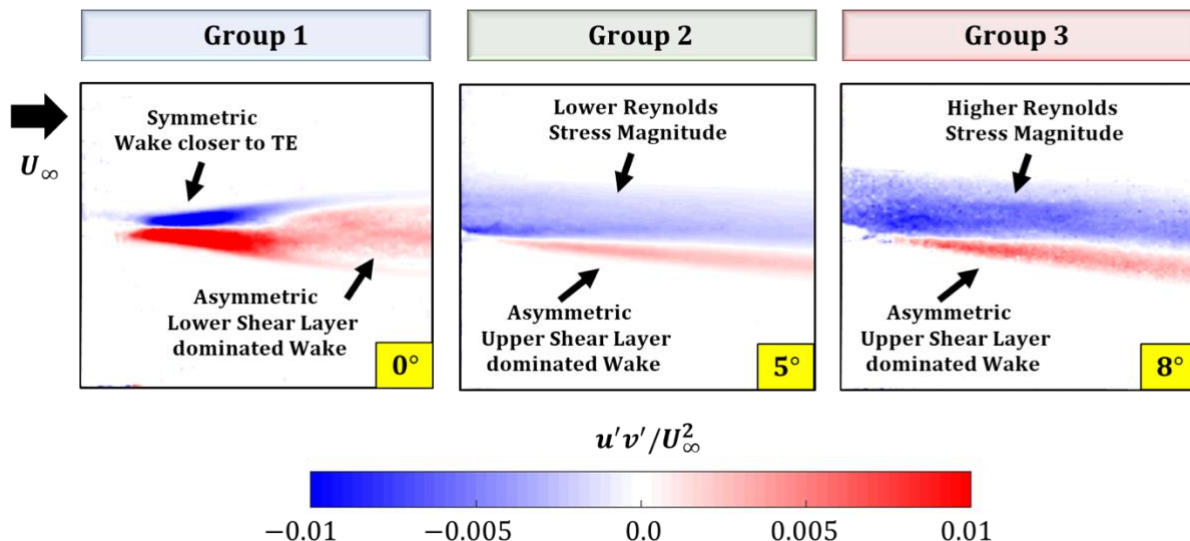


Figure 18 Normalized Reynolds shear stress, $(u'v')/U_\infty^2$, contours at all angles of attack.

Figure 18 contains contours of the normalized Reynolds shear stress, $\frac{u'v'}{U_\infty^2}$, for each angle of attack. A relatively large asymmetry is apparent between the components of the wake free shear layer suspected to be created by the upper (blue band) and lower (red band) surfaces of the airfoil. This is particularly for angles of attack above 1 degree. The lower

angles of attack, -2° and 0° , have more symmetric distribution, likely due to being near the zero-lift angle of attack which is around -1° . Normalized Reynolds stress magnitude can be seen to decrease over the upper surface beyond 2° angle of attack. This has been seen in the other quantities examined in this study as well: a relatively high magnitude around 2° , followed by a decrease up to around the $(L/D)_{\text{Max}}$ angle of attack. Figure 18 also show an increase in Reynolds stress magnitude at 8° which is likely due to onset of flow separation.

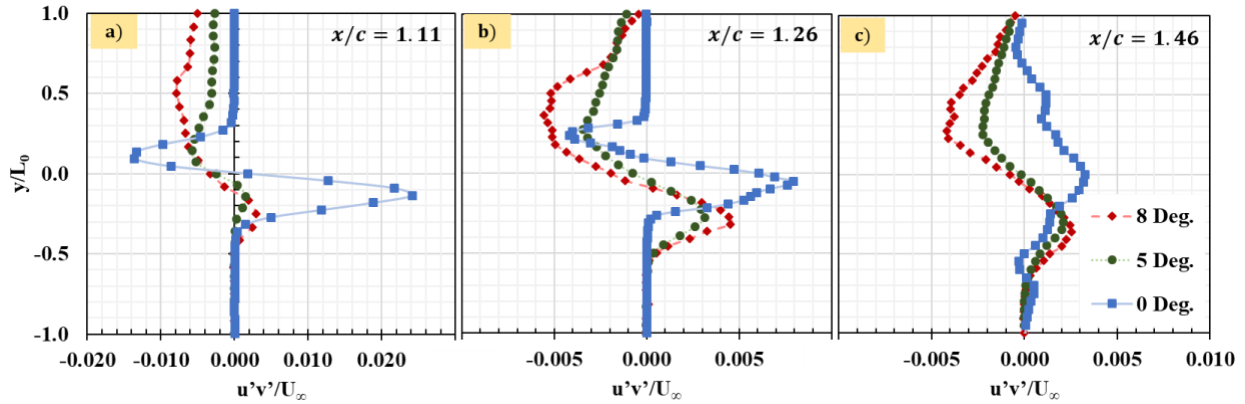


Figure 19 Comparison of normalized Reynolds shear stress at Locations (a) 1 (trailing edge), (b) 2 ($\frac{1}{4}$ chord downstream), (c) 3 ($\frac{1}{2}$ chord downstream).

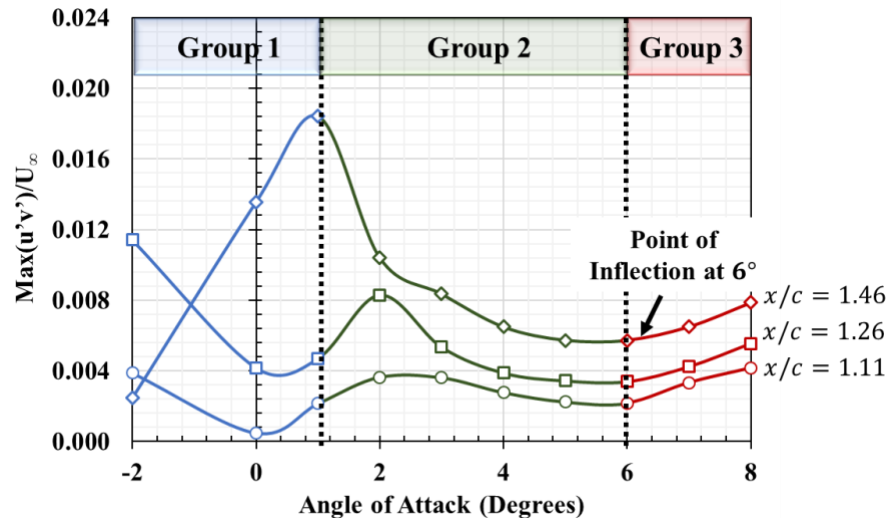


Figure 20 Plot of normalized peak Reynolds shear stress at all angles of attack for all locations along the upper surface.

To more closely study the evolution of the Reynolds shear stress with angle of attack and downstream distance, vertical profile slices were taken at the trailing edge, $\frac{1}{4}$ chord downstream, and $\frac{1}{2}$ chord downstream and were subsequently plotted for all angles of attack. These plots are shown below in Figure 19.

As before, a large spike in the Reynolds shear stress at 0° angle of attack is present. This is attributed to the strong vortex shedding at this α , as was previously discussed. These profiles also highlight the strong asymmetry present between the upper and lower airfoil surface shear layer components. Asymmetry is particularly clear at the trailing edge (Figure 19a) while the distribution becomes more and more symmetric with downstream distance. At lower angles of attack (Group 1), the distribution is still asymmetric, but the local maximum is found both in the upper and lower surface. However, this changes as the angle of attack is increased. As α increases, the local maximum is found on *either* the upper or lower surface: not both. Both upper and lower surface undergoes similar changes. To better understand the behavior of the Reynolds stress around this angle of attack, the peak magnitudes along the upper surface were plotted against angle of attack in Figure 20.

The variation in peak Reynolds shear stress along the upper surface again illustrates what could be interpreted as the presence of three different groupings. As with the other flow properties, the lower angles of attack (Group 1) possess a higher overall unsteadiness. This is attributed to the periodic vortex shedding identified at these α 's through

the temporal evolution of vorticity. Following from 1°, Group 2's values tend to decrease in magnitude with angle of attack. Again, the maximum aerodynamic efficiency condition appears to be a local minimum. A minimum which marks the transition from Group 2 to Group 3. This final angle of attack grouping again shows an increase in magnitude from 6° to 8° angle of attack. These trends are present along both surfaces of the SD7003 wall-to-wall model.

V. Conclusions

Experimental investigation into the SD7003's near wake flow properties as a function of angle of attack, suggests the possibility of three different wake signature groupings.

- Group 1 consists of -2°, 0°, and 1° angles of attack and is heavily identified by periodic vortex shedding. Coherent structures were identified through studying the temporal evolution of vorticity and were found to exist only within this grouping. Additionally, high peak magnitudes of momentum deficit, U_{RMS} , V_{RMS} , and Reynolds shear stress appear to be indicative of these low angles of attack.
- Group 2, which includes 2°, 3°, 4°, and 5° angles of attack, is characterized by a change in magnitude and gradient of flow parameters and decrease in peak magnitude for the mean and fluctuating flow quantities. It also happens that this overall decreasing trend stops in the vicinity of the maximum aerodynamic efficiency condition of 5°.
- Beyond $(L/D)_{Max}$ is the grouping of angles of attack comprising Group 3: 6°, 7°, and 8°. This final grouping tends to have an inflection point in most of the flow parameters which is particularly evident in the fluctuating quantities. Neither Group 2 nor Group 3 have periodic vortex shedding in the $\frac{1}{2}$ chord length that was investigated.

These conclusions indicate that the near wake properties reflect similar trends as the maximum aerodynamic efficiency. These near wake properties indicate inherent changes in the wake signature before and after the condition of maximum aerodynamic efficiency. The wake signature change and reduction in vortex shedding from Group 1 to Group 2 can be attributed to the formation of a laminar separation bubble on the upper surface of the SD7003 airfoil. The wake signature results when taken into consideration along with the boundary layer displacement thickness does imply onset of flow separation just after maximum aerodynamic efficiency condition.

VI. Future Work

The flow dynamics observed in Group 1 associated with strong vortex shedding could occur due to the onset of a Kelvin-Helmholtz instability in the boundary layer. This could be verified using on-body boundary layer PIV experiments. However, the sudden disappearance of vortex shedding at 2° angle of attack (with the change in wake structure occurring at just 1° increment) warrants further investigations. If the mechanism for the sudden reduction in vortex shedding can be understood through unsteady pressure and on-body PIV analysis, it could be applied to several applications where vortex shedding results in aeroacoustic perturbation (noise) and potentially damaging structural vibrations.

VII. Acknowledgments

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